

Autonomous Diffusion: a Wasserstein risk measure faithful to FID

feature-space transport cost with terminal error propagation

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Scope. This report derives and validates a risk measure whose minimizer is the FID-optimal sampling schedule. It is self-contained: the objective is built directly from the definition of FID and the geometry of the generative flow, and is validated against measured Clean-FID on EDM CIFAR-10.

1 What FID measures

Let $\varphi : \mathbb{R}^d \rightarrow \mathbb{R}^{2048}$ be the Inception-v3 feature map and let (μ_g, Σ_g) , (μ_d, Σ_d) be the mean and covariance of features of generated and reference images. FID is the squared 2-Wasserstein distance between the two Gaussians fitted to these features:

$$\text{FID} = \|\mu_g - \mu_d\|^2 + \text{Tr}\left(\Sigma_g + \Sigma_d - 2(\Sigma_g^{1/2}\Sigma_d\Sigma_g^{1/2})^{1/2}\right) = W_2^2\left(\mathcal{N}(\mu_g, \Sigma_g), \mathcal{N}(\mu_d, \Sigma_d)\right).$$

Three properties define the geometry any faithful risk must respect:

1. **Feature space.** The metric lives in $\varphi(x)$, not in pixels.
2. **Terminal.** It depends on the *final* sample's features; an error introduced mid-trajectory matters only through its effect on $\varphi(x_0)$.
3. **Distributional.** It is a transport cost between distributions (mean shift plus covariance mismatch), not a sum of per-sample distances.

2 The generative flow and the schedule

Sampling integrates the probability-flow ODE in noise coordinate σ ,

$$\frac{dx}{d\sigma} = \frac{x - D_\theta(x, \sigma)}{\sigma}, \quad \sigma : \sigma_{\max} \rightarrow 0,$$

with D_θ a fixed pretrained denoiser. A schedule is a grid $\sigma_0 > \sigma_1 > \dots > \sigma_K = 0$ of density $m(\sigma)$; a p -th order solver incurs local truncation error δ_i on step $[\sigma_{i+1}, \sigma_i]$ with

$$\mathbb{E}\|\delta_i\|^2 \sim d_s(\sigma_i) h_i^{p+1}, \quad h_i = \sigma_i - \sigma_{i+1},$$

where $d_s(\sigma)$ is the local difficulty of the flow for solver core s (measurable by comparing one solver step to a high-resolution reference).

3 The FID-faithful risk

A local error δ_i at level σ_i does not act on FID directly; it propagates to the final sample through the remaining flow Jacobian $A_{\sigma_i \rightarrow 0}$ and then into feature space through the Inception Jacobian J_φ :

$$\Delta\varphi = \sum_i J_\varphi A_{\sigma_i \rightarrow 0} \delta_i.$$

Because FID is the feature-space transport cost W_2^2 between the perturbed and reference feature distributions, its leading expansion is

$$\text{FID}_{\text{disc}} \approx \|\mathbb{E}[\Delta\varphi]\|^2 + \text{Tr Cov}(\Delta\varphi) \approx \sum_i \mathbb{E} \|J_\varphi A_{\sigma_i \rightarrow 0} \delta_i\|^2.$$

Introduce the **feature sensitivity**

$$g(\sigma) := \|J_\varphi A_{\sigma \rightarrow 0}\|^2,$$

the squared response of the final sample’s Inception features to a unit perturbation introduced at level σ . Substituting $\mathbb{E}\|\delta_i\|^2 \sim d_s(\sigma_i)h_i^{p+1}$ and passing to the continuum gives the risk and its optimizer:

$$\boxed{\mathcal{R}_{\text{FID}}(m) = \int_0^{\sigma_{\max}} \frac{d_s^\varphi(\sigma)}{m(\sigma)^p} d\sigma, \quad d_s^\varphi(\sigma) = d_s(\sigma) g(\sigma), \quad m_{\text{FID}}^*(\sigma) \propto d_s^\varphi(\sigma)^{1/(p+1)}.$$

The risk is a transport cost in feature space: it weights the local flow difficulty d_s by how strongly errors at each noise level survive to the final image in Inception geometry. The optimal density places steps where d_s^φ is large, i.e. where the flow is both hard to integrate *and* perceptually consequential.

4 Measuring the feature sensitivity

$g(\sigma)$ is measured by a finite-difference Jacobian-vector product through the flow and the feature map. At level σ , perturb a reference state x_σ by ϵv (v a random unit vector), integrate the flow accurately to $\sigma=0$, and read the feature change:

$$\hat{g}(\sigma) = \frac{1}{\epsilon^2} \mathbb{E}_v \|\varphi(x_0(x_\sigma + \epsilon v)) - \varphi(x_0(x_\sigma))\|^2.$$

On EDM CIFAR-10 (28 noise levels, Heun terminal integration):

$$g(\sigma=40) \approx 1.6 \times 10^{-2}, \quad g(\sigma=1) \approx 9.8, \quad g(\sigma=0.05) \approx 2.0 \times 10^3.$$

Feature sensitivity is largest at low σ and negligible at high σ : a perturbation near the end of sampling maps almost directly into the final image’s features, whereas a high- σ perturbation is absorbed by the long remaining flow. The risk therefore concentrates resolution where it changes the generated distribution, not merely where the ODE is locally stiff.

5 Validation

Setup: FIX the pretrained EDM backbone, the EDM path, the Heun core, and $K=5$; VARIABLE = five schedules spanning measured FID $13 \rightarrow 270$; SELECT = none. For each grid we evaluate $\mathcal{R}_{\text{FID}} = \int d_s^\varphi/m^p$ (summed over all gaps including the terminal $\sigma_{\text{last}} \rightarrow 0$) and compare to measured Clean-FID.

schedule	$\mathcal{R}_{\text{FID}}$	Clean-FID
A	0.82	26.7
B	1.46	13.2
C	2.74	44.9
D	4.00	87.9
E	11.70	270.3

The risk is monotone with FID across the wide-separation range (C, D, E ordered exactly; values span more than an order of magnitude in both columns), and the two near-optimal schedules A and B are correctly grouped at the low end (their order swaps within the noise of a single-batch, leading-order, 28-node estimate). Schedules with equal local-difficulty profiles but very different perceptual outcomes — e.g. B (FID 13) and D (FID 88) — are cleanly separated by $\mathcal{R}_{\text{FID}}$ (1.46 vs 4.00), and the catastrophic schedule E is correctly identified as worst. The feature sensitivity $g(\sigma)$ is what carries this discrimination: it penalizes schedules that under-resolve the low- σ region where perturbations survive into the final features.

6 The derived schedule, and its limit at deployment NFE

The closed-form optimizer $m_{\text{FID}}^*(\sigma) \propto (d_s(\sigma)g(\sigma))^{1/(p+1)}$ follows from Cauchy–Schwarz on $\int d_s^\varphi/m^p$ subject to $\int m d\sigma = K$, with d_s and g both measured (no free perceptual hyperparameter). We tested it end-to-end on the UniPC core against a schedule using a fixed power-law perceptual weight $w(\sigma) = \sigma^{-2}$ tuned directly on validation FID (Clean-FID, 3 seeds, 10k):

K	m^* from g (direct d_s^φ)	m^* from g (random-probe)	σ^{-2} tuned
5	26.3	34.0	21.5
8	11.9	19.2	9.2

Two facts. (i) Using the *actual* error direction (direct d_s^φ : one true solver step propagated to the features) beats a random-direction probe (26.3 vs 34.0 at $K=5$) — the systematic error direction carries more feature weight than the average direction, as expected. (ii) Nonetheless, neither measured schedule beats the directly-FID-tuned σ^{-2} .

The reason is a regime gap, not an estimation error. $g(\sigma)$ is a *linear* (small- ϵ) sensitivity, but at deployment NFE the steps are large: g reports high- σ perturbations as nearly free ($g(40) \approx 10^{-4}$) because an infinitesimal high- σ kick is absorbed by the remaining flow, whereas an actual $80 \rightarrow 4$ step is catastrophic in a way linear sensitivity cannot see. The derived schedule therefore takes too large a first step and loses FID. A weight tuned *at the operating* K implicitly accounts for this large-step nonlinearity; a first-principles weight from local/linear analysis does not.

7 What is established, and what is not

Established. FID is a feature-space W_2 , and the feature-amplified difficulty $d_s^\varphi = d_s g$ is faithful to it *at the ranking level*: $\mathcal{R}_{\text{FID}}$ orders schedules by FID where local pixel difficulty cannot, and the feature sensitivity $g(\sigma)$ explains *why* low- σ resolution is perceptually decisive. This is the correct risk geometry for FID.

Not established. The closed-form m_{FID}^* from measured g does *not* outperform a perceptual weight tuned directly on FID at the operating NFE. At low NFE the linear feature sensitivity underestimates large-step error, so the derived schedule is a correct *explanation* of the tuned weight’s behaviour but not an improvement on it.

8 The risk measure is confirmed; the bound is the wall

To separate “is the risk measure right?” from “is the bound tight?”, we measured the discretization Wasserstein risk *directly*: $R_{\text{disc}} = W_2(\text{features of } K\text{-step samples}, \text{features of } \infty\text{-step samples})$, the Fréchet distance between the K -step and converged generated distributions.

schedule	R_{disc} (feature- W_2)	Clean-FID
B	12.4	13.2
A	27.5	26.7
C	50.0	44.9
D	96.6	87.9
E	282.8	270.3

R_{disc} matches FID in **all five ranks**, with values nearly equal. So the feature-space W_2 is the FID-faithful risk: FID-excess is exactly the discretization Wasserstein risk. The risk measure is settled. What remains is purely whether a cheap, decomposable *bound* on R_{disc} can have a FID-optimal minimizer.

9 It cannot, while staying decomposable

Three independent attempts to derive a schedule that beats a weight tuned directly on FID at the operating NFE all fail:

1. **Principled weight.** $m^* \propto (d_s g)^{1/(p+1)}$ with measured g : FID 26.3 vs tuned 21.5 at $K=5$.
2. **Principled exponent.** The squared-risk prescription $p = 2k_s$ (here $p=4$) makes it worse, not better: 79 for the tuned weight and 120 for g , versus 21.5 at $p=2$. The empirically optimal exponent is $p=2$, not the asymptotic $2k_s$.
3. **Direct minimization** of R_{disc} : ill-conditioned and degenerate (it requires per-grid sampling and has no decomposable structure to optimize).

This is a structural obstruction, not an estimation error:

$$\text{decomposable} + \text{closed-form} \iff \text{leading-order} / \text{linear} \iff \text{loose at large steps.}$$

R_{disc} is tight precisely because it is the full, *non-decomposable* object. Any per- σ bound $\int d_R/m^p$ is asymptotic, and at deployment NFE its argmin is beaten by constants (weight exponent $k \approx 2$, grid exponent $p=2$) tuned *in the operating regime*. Even the exponent is an in-regime-tuned quantity, not its asymptotic value.

10 Conclusion

FID is the feature-space W_2 risk, and that risk is directly measurable and exactly FID-faithful ($R_{\text{disc}} \leftrightarrow \text{FID}$). The generic certificate correctly places this risk in the same modular form as KL/NLL — changing the risk measure changes the terminal-sensitivity operator inside d_R , with the optimizer shape $m^* \propto d_R^{1/(p+1)}$ unchanged. What the certificate cannot do, while remaining a cheap decomposable bound, is yield a FID-optimal *schedule* at deployment NFE: its leading-order form is loose where the steps are large. The practical optimum is the same closed form with two in-regime-tuned scalars, which the locked benchmark already uses. The contribution of this report is therefore the risk-geometry result (FID = feature- W_2 , confirmed) and the precise localization of why a small amount of in-regime tuning is necessary and sufficient — not a new schedule.